# Area from a base and height

## Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

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Lesson focus: Find area by matching a base with its perpendicular height.

R1. Find $2\frac{1}{2} \div \frac{3}{4}$.

R2. Three notebooks cost $2.45 each. What is the total cost?

R3. Find $3^2$. Write the repeated factors.

R4. Find $4 \div \frac{2}{3}$.

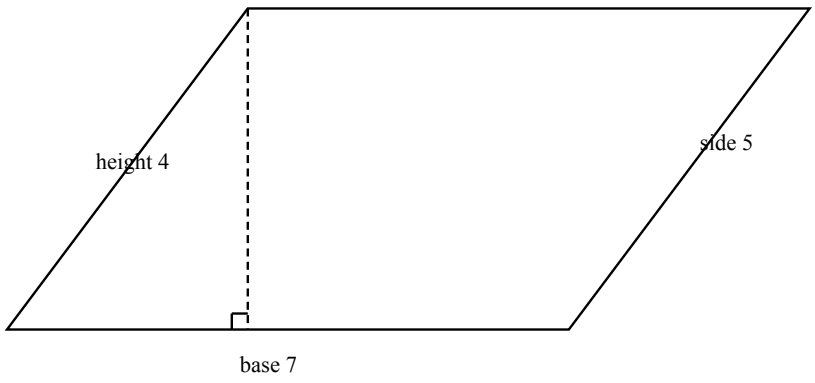
P1. A rectangle is 6 cm long and 2.5 cm wide. Find its area and perimeter.

P2. A parallelogram has base 7 cm, height 4 cm, and slanted side 5 cm. Cut and move the end triangle. Find its area.

P3. A right triangle has perpendicular legs of 6 cm and 8 cm. Find its area. Use either leg as the base.

P4. An isosceles triangle has sides 13, 13, and 10 cm. Its height to the 10 cm base is 12 cm. Find the area.

## Lesson model



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### Practice and compare methods

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P5. A triangle has vertices A(0, 0), B(4, 0), and C(6, 3), measured in cm. The height meets the extended base at (6, 0). Find its area.

P6. A triangle has base 5 cm and matching height 2.4 cm. Find its area. Would base × slanted side always work?

F. A triangle has base 8 cm and height 2.5 cm. Complete: $A = \frac{1}{2} \times 8 \times __ = __$.

D. Theo uses a triangle's base 9 cm and height 4 cm. He writes 9 × 4 = 36 square cm. What is missing?

X1. A rectangle has length 6 cm and width 3 cm. Find area and perimeter.

X2. A parallelogram has base 4 cm, height 3 cm, and slanted side 5 cm. Find area and name the unused length.

X3. A right triangle has perpendicular legs 8 cm and 3 cm. Find area using either leg as base.

X4. A triangle has area 12 square cm and base 8 cm. Find its matching height.

### Worked example - compare with the slide model

A parallelogram has base 6 cm and height 3 cm.

Cut a triangle from one end and move it to the other.

The new rectangle has the same area, base 6 and height 3.

Area = 6 × 3 = 18 square cm.

**Result: 18 square cm**

# Area from a base and height

## Challenge, explain and check

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C1. A parallelogram garden has base 12 m, perpendicular height 5 m, and slanted side 7 m. A triangular pond with base 6 m and height 4 m lies inside it. Find the planted area. Explain why using 7 m as the garden height changes the result incorrectly.

C2. A triangle has base 10 cm and height 12 cm. Design a rectangle of the same area with one side 5 cm. Compare their areas, then explain whether equal area guarantees equal perimeter. Use the triangle sides 10, 13, 13 cm for the perimeter comparison.

## Finish the lesson

S. For this base, the matching height is __ because __.

E. A triangle has sides 10, 10, and 12 cm. The height to the 12 cm base is 8 cm. Find its area and name the length you used as height.